Algebra and Discrete Mathematics (ADM)

Tutorial 8 Eulerian tours

Lecturer: doc. Xiaolu Hou, PhD.
xiaolu.hou@stuba.sk

Graph

Definition

A *graph* consists of two sets: $V(G)$, called the *vertex set*, and $E(G)$, called the edge set. An *edge*, denoted xy, is an unordered pair of vertices $x, y \in V(G)$, where x and y may be the same.

Theorem (Handshaking Lemma)

Let $G = (V, E)$ be a finite graph. Suppose $V = \{v_1, v_2, \ldots, v_n\}$, then

$$\deg(v_1) + \deg(v_2) + \cdots + \deg(v_n) = 2|E|$$

Draw a graph

Question

Draw a graph with 6 vertices such that their degrees are

$$1,\ 2,\ 3,\ 4,\ 5,\ 2$$

We have

$$\sum_{v} \deg(v) = 1 + 2 + 3 + 4 + 5 + 2 = 17,$$

which is odd. Hence, such a graph does not exist.

Draw a graph

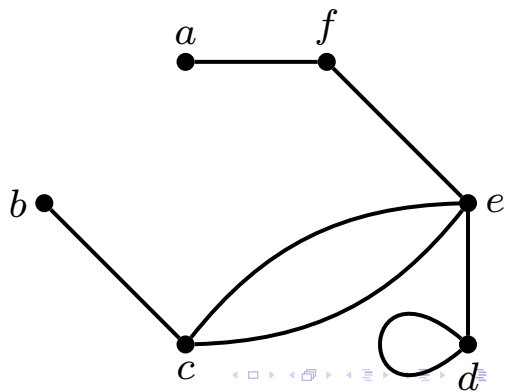
Question

Draw a graph with 6 vertices such that their degrees are

$$1,\ 1,\ 3,\ 3,\ 4,\ 2$$

$\deg(a) = 1, \quad \deg(b) = 1, \quad \deg(c) = 3,$

$\deg(d) = 3, \quad \deg(e) = 4, \quad \deg(f) = 2,$



Draw a graph

Question

Construct a graph to represent a club with 13 members, where each member is acquainted with exactly 5 other members.

The sum of degrees is given by

$$13 \times 5 = 65$$

which is odd. Hence, such a graph does not exist.

Non-existence of certain regular graphs

Statement

Prove that a k-regular graph, where k is odd, with an odd number of vertices does not exist.

Proof.

If a graph has n vertices and is k-regular, then the sum of all vertex degrees is

$$nk.$$

Since both n and k are odd, the product nk is odd.
But by the Handshaking Lemma, the sum of the degrees of all vertices must be even.
This is a contradiction, so such a graph cannot exist. □

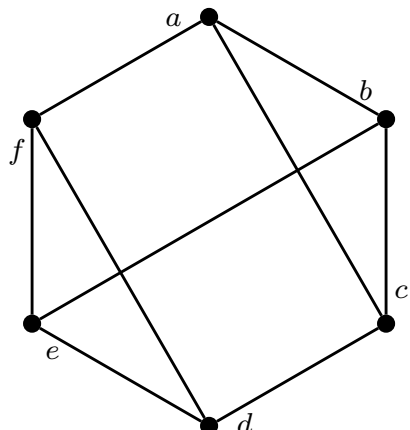
Draw a graph

Question

Draw a graph with 6 vertices, each of degree 3

$$\frac{6 \times 3}{2} = 9$$

so the graph must have 9 edges.



Draw a graph

Question

Draw a graph with 7 vertices, each of degree 3

$$7 \times 3 = 21,$$

which is odd. Hence, such a graph does not exist.

Draw a graph

Question

Draw a graph with 4 vertices, each of degree 1



Note that this graph is not connected.

Draw a graph

Question

Draw a graph with 6 edges, 4 vertices with degrees

$$1,\ 2,\ 3,\ 4$$

It must hold that

$$2|E| = \sum_{v \in V(G)} \deg(v)$$

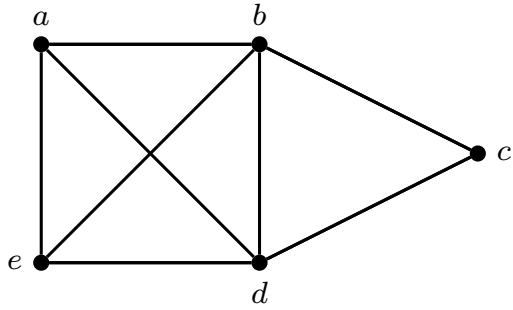
However

$$2|E| = 2 \times 6 = 12$$

$$\sum_{v \in V(G)} \deg(v) = 1 + 2 + 3 + 4 = 10$$

Hence, such a graph does not exist.

Touring a graph



- Path: does not repeat vertices
  - Example: $a,\ b,\ c,\ d,\ e$
- Trail: does not repeat edges
  - Example: $a,\ b,\ e,\ a,\ d,\ c$
- Cycle: a closed path
  - Example: $a,\ b,\ c,\ d,\ e,\ a$
- Circuit: a closed trail
  - Example: $a,\ b,\ c,\ d,\ b,\ e,\ a$

Eulerian circuit

Definition

Let G be a graph. An *Eulerian circuit* is a circuit that uses every edge of G exactly once. An *Eulerian trail* is an open[a] trail that uses every edge of G exactly once. If a *connected* graph G contains an Eulerian circuit, then G is called *Eulerian*. If a *connected* graph G contains an Eulerian trail but not an Eulerian circuit, then G is called *semi-Eulerian*.

[a] not a circuit

Note

A graph may contain an Eulerian circuit (or trail) even if it is not connected, provided that all vertices with nonzero degree belong to the same connected component. In this course, we require Eulerian and semi-Eulerian graphs to be connected.

Eulerian and semi-Eulerian

Theorem

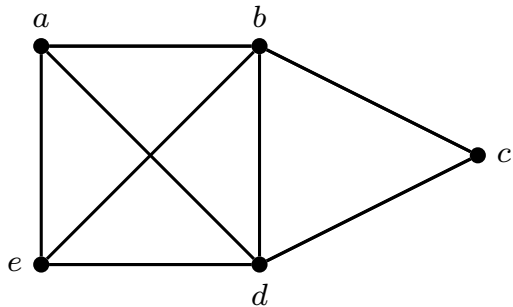
A graph G is Eulerian if and only if

- *G is connected and*
- *every vertex has an even degree*

A graph G is semi-Eulerian if and only if

- *G is connected and*
- *exactly two vertices have odd degree*

Semi-Eulerian graph



$$\deg(a) = 3, \quad \deg(b) = 4, \quad \deg(c) = 2, \quad \deg(d) = 4, \quad \deg(e) = 3$$

The graph is semi-Eulerian

Fleury's Algorithm

1. Choose a starting vertex:
   - If the graph G has no vertices of odd degree, start from any vertex
   - If G has exactly two vertices of odd degree, start from one of them
2. Choose an edge to traverse: Select an edge vw incident to the current vertex v such that, after removing it, v and w are still connected, unless no other choice is available.
3. Traverse and remove the edge: Move to the other endpoint of the selected edge, then delete the edge from the graph. The new vertex becomes the current vertex.
4. Repeat Steps 2 and 3 until there are no more edges left in the graph.

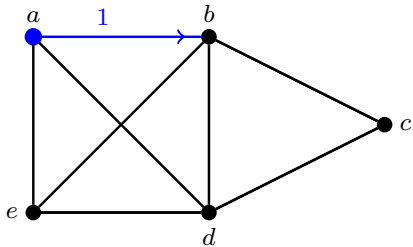
Output:

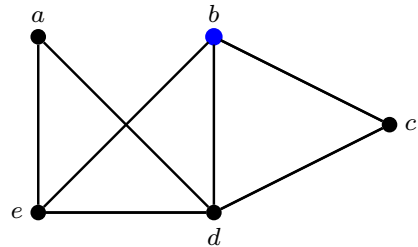
- If G originally had no odd-degree vertices, Eulerian circuit
- If G had exactly two odd-degree vertices, Eulerian trail

Practical implementation: Maintain two copies of the graph:

- One for tracking the Eulerian trail or circuit.
- Another for dynamically removing edges

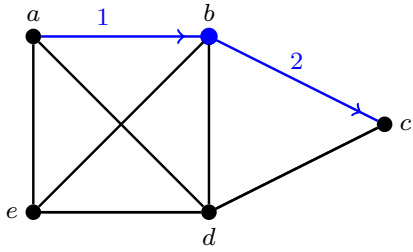
Fleury's Algorithm

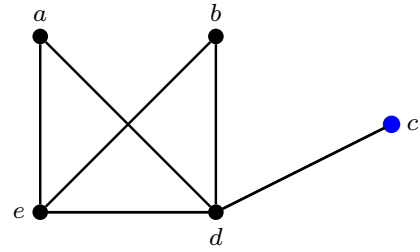




- Step 1. Start at vertex a
- Step 2. Choose edge ab
- Step 3. Travel to vertex b and delete edge ab. Current vertex: b.

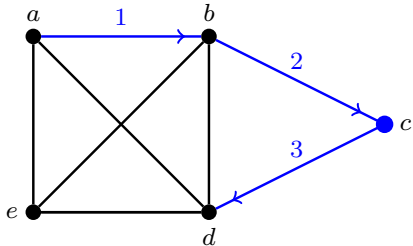
Fleury's Algorithm

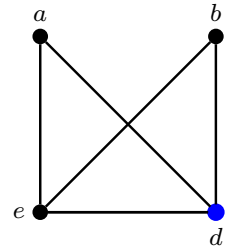




- Step 2. Choose edge bc
- Step 3. Travel to vertex c and delete edge bc. Current vertex: c.

Fleury's Algorithm

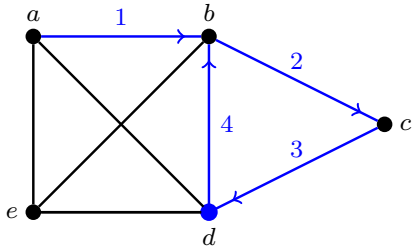


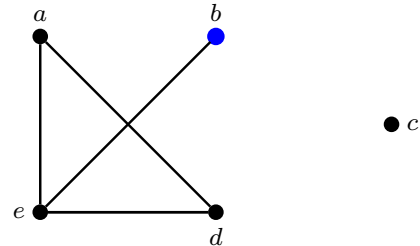




- Step 2. The only available choice is edge cd.
- Step 3. Travel to vertex d and delete edge cd. Current vertex: d.

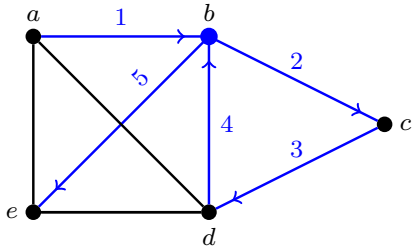
Fleury's Algorithm

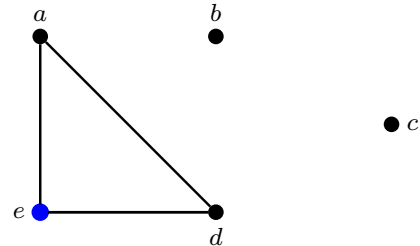




- Step 2. Choose edge db
- Step 3. Travel to vertex b and delete edge db. Current vertex: b.

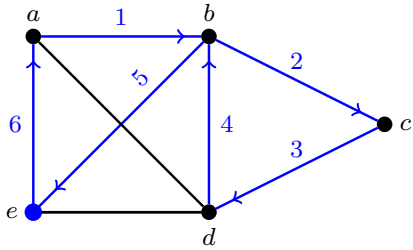
Fleury's Algorithm

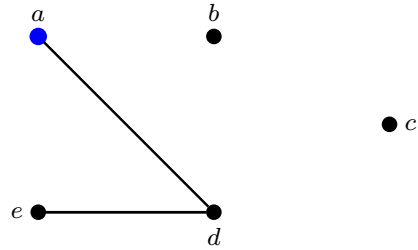




- Step 2. The only available choice is be
- Step 3. Travel to vertex e and delete edge be. Current vertex: e.

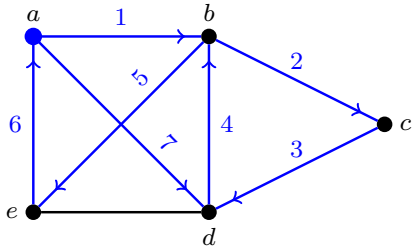
Fleury's Algorithm

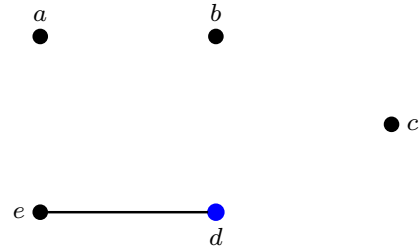




- Step 2. Choose ea
- Step 3. Travel to vertex a and delete edge ea. Current vertex: a.

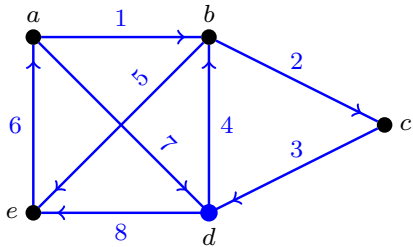
Fleury's Algorithm

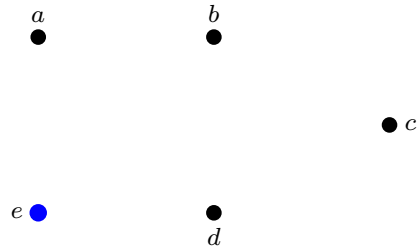




- Step 2. The only available choice is ad
- Step 3. Travel to vertex d and delete edge ad. Current vertex: d.

Fleury's Algorithm





- Step 2. The only available choice is de
- Step 3. Travel to vertex e and delete edge de.
- We have obtained an Eulerian trail

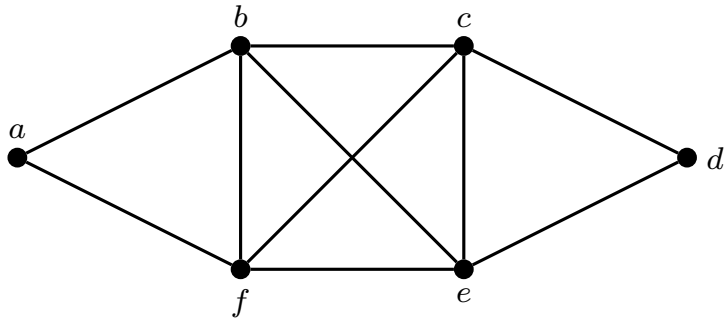
Hierholzer's Algorithm

- Find an Eulerian circuit
- Input: an Eulerian graph G – connected, all vertices are even

Hierholzer's Algorithm

1. Choose a starting vertex, say v, and find an initial circuit C starting at v.
2. If there exists a vertex x in the current circuit C that has unused edges, form a new circuit C' starting and ending at x, using only unused edges.
3. Merge the newly found circuit C' into the existing circuit C, updating C accordingly.
4. Repeat steps 2 and 3 until all edges of G have been traversed.

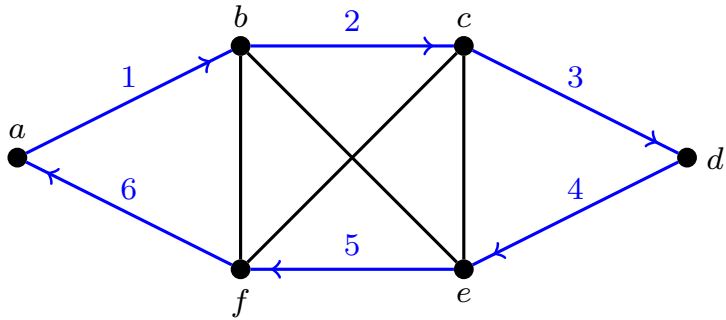
Hierholzer's Algorithm



$\deg(a) = 2, \quad \deg(b) = 4, \quad \deg(c) = 4, \quad \deg(d) = 2, \quad \deg(e) = 4, \quad \deg(f) = 4$

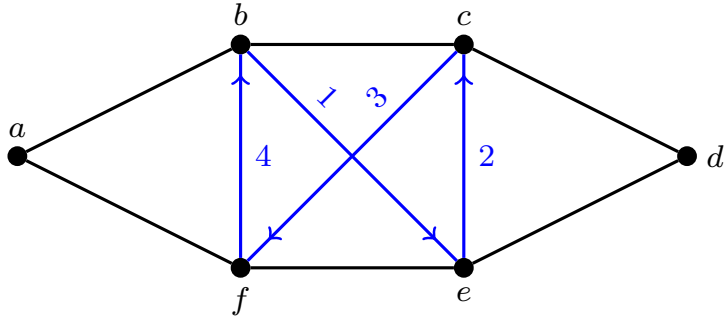
The graph is Eulerian.

Hierholzer's Algorithm



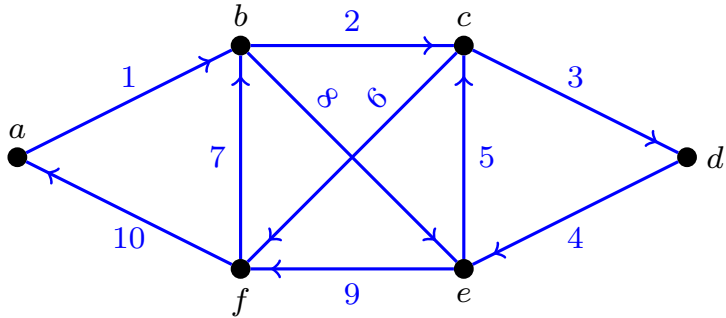
- Step 1. Choose vertex a and find a circuit starting at a.

Hierholzer's Algorithm



- Step 2. Since $\deg(b) = 4$, two unused edges remain incident to b, so we can find a second circuit starting at b.

Hierholzer's Algorithm



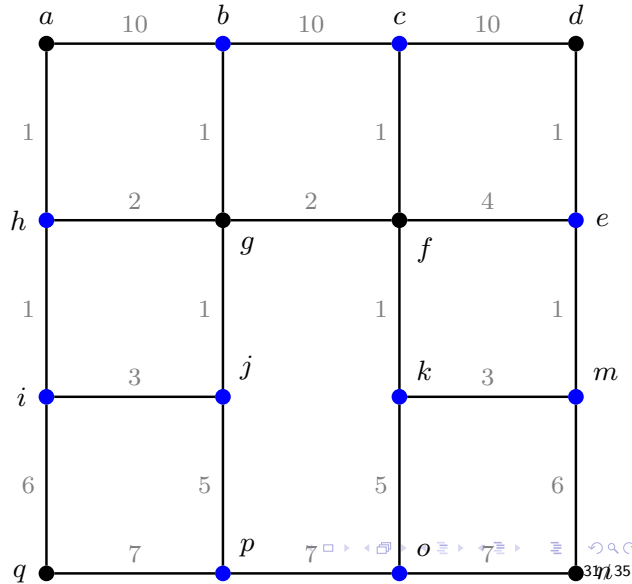
- Step 3. Combine the new circuit with the existing one.
- We have obtained an Eulerian circuit.

Chinese Postman Problem

- The *Chinese Postman Problem* is to find a closed walk of minimum total weight that traverses every edge of a connected, undirected, weighted graph at least once.
- It can be solved by finding an Eulerization with minimum added weight, followed by an Eulerian circuit in the resulting graph.
- The name originates not from anything particular about postmen in China, but rather from the mathematician who first proposed the problem —the Chinese mathematician Mei-Ku Kwan (管梅谷) in 1962

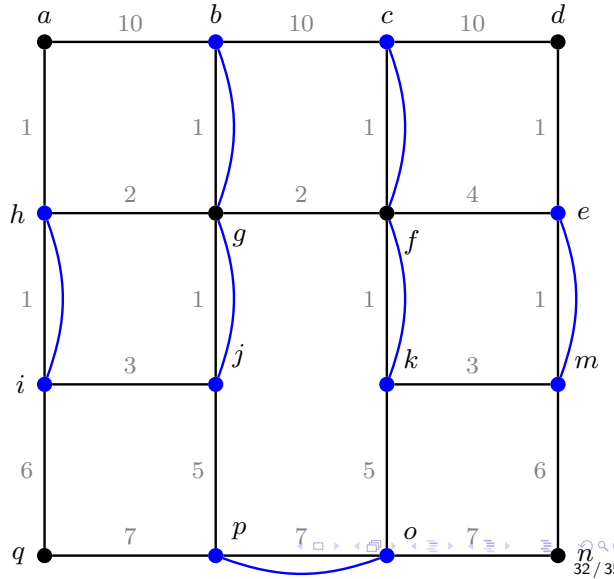
Chinese Postman Problem

- Find an optimal Eulerization, where the edge weights represent cost.
- We first attempt to pair vertices along paths using edges of weight 1 as much as possible
- We are able to do this for all vertices except o and p, and we duplicate edge op



Chinese Postman Problem

- The graph has 10 odd vertices, so at least 5 edge duplications are needed.
- If we avoid duplicating op, then o and p each require a duplicated incident edge of weight at least 5. The other duplications contribute at least 3, so the total added weight is at least 13.
- If we duplicate op, its weight is 7. Among the remaining 8 odd vertices, only hi and em are adjacent via edges of weight 1. Thus, at least two pairs require paths of weight at least 2, giving an additional weight of at least $1 + 1 + 2 + 2 = 6$.
- Therefore, the Eulerization shown, with total added weight 13, is optimal.



Week 10 – Next week

- **Lecture:** As usual
- **Tutorial:** As usual

Week 11 – Make-up Quiz 1

- **Lecture:** 9:00–9:50
  - Discussion of tutorial 10
- **Make-up Quiz 1**
  - Time: **10:00–12:20**
- **No regular tutorial this week**

Week 12 – Make-up Quiz 2

- **Lecture:** 9:00–10:50
- **Make-up Quiz 2**
  - Time: **11:00–13:20**
- **No regular tutorial this week**